13.1 ECDIS should be provided with means for either automatically or manually carrying out onboard tests of major functions. In case of a failure, the test should display information to indicate which module is at fault.

13.2 ECDIS should provide a suitable warning or indication of system malfunction.

14 BACKUP ARRANGEMENTS

Adequate backup arrangements should be provided to ensure safe navigation in case of an ECDIS failure; see appendix 6.

.1 Facilities enabling a safe takeover of the ECDIS functions should be provided in order to ensure that an ECDIS failure does not develop into a critical situation.

SOURCE: MSC.530(106).pdf
URL:
https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.530(106).pdf

provided in order to ensure that an ECDIS failure does not develop into a critical situation.

.2 A backup arrangement should provide means of safe navigation for the remaining part of a voyage in the case of an ECDIS failure.

MODULE C INTERFACING AND INTEGRATION

15 CONNECTIONS WITH OTHER EQUIPMENT7

15.1 ECDIS should not degrade the performance of any equipment providing sensor inputs. Nor should the connection of optional equipment degrade the performance of ECDIS below this standard.

15.2 ECDIS should be connected to the ship's position fixing system, to the gyro compass and the speed and distance measuring device. For ships not fitted with a gyro compass, ECDIS should be connected to a marine transmitting heading device.

15.3 ECDIS may provide a means to supply system database information to external equipment.

16 POWER SUPPLY

16.1 It should be possible to operate ECDIS and all equipment necessary for its normal

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equipment.

16 POWER SUPPLY

16.1 It should be possible to operate ECDIS and all equipment necessary for its normal functioning when supplied by an emergency source of electrical power in accordance with the

appropriate requirements of SOLAS chapter II-1.

16.2 Changing from one source of power supply to another or any interruption of the supply for a period of up to 45 seconds should not require the equipment to be manually re-initialized.

7 Publication IEC 61162.

APPENDIX 1

REFERENCE DOCUMENTS

The following international organizations have developed technical standards and specifications, as listed below, for use in conjunction with this standard. The latest edition of these documents should be obtained from the organization concerned:

SOURCE: MSC.530(106).pdf
URL:
https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.530(106).pdf
these documents should be obtained from the organization concerned:

INTERNATIONAL MARITIME ORGANIZATION (IMO)

Address: International Maritime Organization Phone: +44 207 735 76 11
4 Albert Embankment Fax: +44 207 587 32 10
London SE1 7SR Email: info@imo.org
United Kingdom Web: http://www.imo.org

Publications

Resolution MSC.191(79), as amended by resolution MSC.466(101), on Performance standards for the presentation of navigation -related information on shipborne navigational displays

Resolution A.694(17) on Recommendations on general requirements for shipborne radio equipment forming part of the Global Maritime Distress and Safety System (GMDSS) and for electronic navigational aids

Resolution MSC.302(87) on Performance standards for bridge alert management

SN.1/Circ.207 on Differences between RCDS and ECDIS

SN.1/Circ.243/Rev.2 on Guidelines fo r the presentation of navigation -related symbols, terms and abbreviations
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SOURCE: MSC.530(106).pdf

URL:
https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.530(106).pdf
SN.1/Circ.243/Rev.2 on Guidelines fo r the presentation of navigation -related symbols, terms and abbreviations

MSC/Circ.982 on Guidelines on ergonomic criteria for bridge equipment and layout

INTERNATIONAL HYDROGRAPHIC ORGANIZATION (IHO)

Address: Directing Committee Phone: +377 93 10 81 00
International Hydrographic Organization
BP 445 Fax: +377 93 10 81 40
MC 98011 Monaco Cedex Email: info@iho.int
Principality of Monaco Web: http://www.iho.int

Publications

IHO Publication S -52, Specifications for Chart Content and Display Aspects of ECDIS

IHO Publication S-52 appendix 1, Guidance on Updating the Electronic Navigational Chart

IHO Publication S-52 appendix 2, Colour and Symbol Specifications for ECDIS

IHO Publication S-32, Hydrographic Dictionary
RESOLUTION MSC.530(106) (adopted on 7 November 2022)
PERFORMANCE STANDARDS FOR ELECTRONIC CHART DISPLAY AND INFORMATION SYSTEMS (ECDIS)
MSC 106/19/Add.1
Annex 22, page 14

SOURCE: MSC.530(106).pdf
URL:
https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.530(106).pdf
PERFORMANCE STANDARDS FOR ELECTRONIC CHART DISPLAY AND INFORMATION SYSTEMS (ECDIS)
MSC 106/19/Add.1
Annex 22, page 14

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IHO Publication S-57, IHO Transfer Standard for Digital Hydrographic Data

IHO Publication S-100, IHO Universal Hydrographic Data Model

IHO Publication S-101 ENC Product Specification

IHO Publication S-98 Data Product Interoperability in S-100 Navigation Systems

IHO Publication S-61, IHO Product specification for Raster Navigational Charts (RNC)

IHO Publication S-63, IHO Data Protection Scheme

IHO Publication M-3, Resolutions of the IHO

https://iho.int/en/standards-in-force

INTERNATIONAL ELECTROTECHNICAL COMMISSION (IEC)

Address: IEC Central Office Phone: +41 22 734 01 /50
3 rue de Varembe Fax: +41 22 733 38 43
PO Box 131
CH-1211 Geneva 20
Switzerland

Publications

IEC Publication 61174, Electronic Chart Display and Information Systems (ECDIS) -
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SOURCE: MSC.530(106).pdf
URL:
https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.530(106).pdf
PO Box 131
CH-1211 Geneva 20
Switzerland

Publications

IEC Publication 61174, Electronic Chart Display and Information Systems (ECDIS) - Operational and Performance Requirements, Method of Testing and Required Test Results.

IEC Publication 60945, General Requirements for Shipborne Radio Equipmen t Forming Part of the Global Maritime Distress and Safety System and Marine Navigational Equipment.